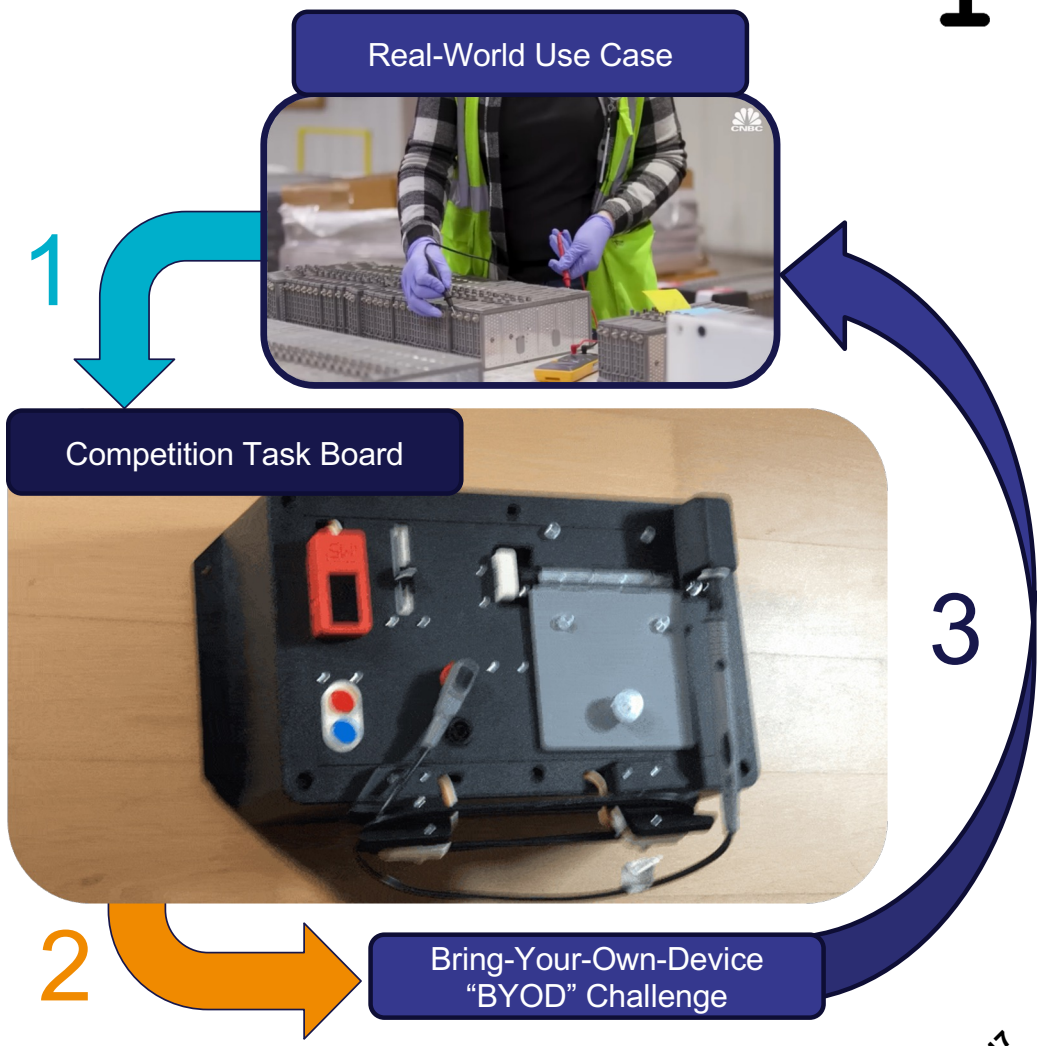
# The Challenge 2023: Using a Multimeter

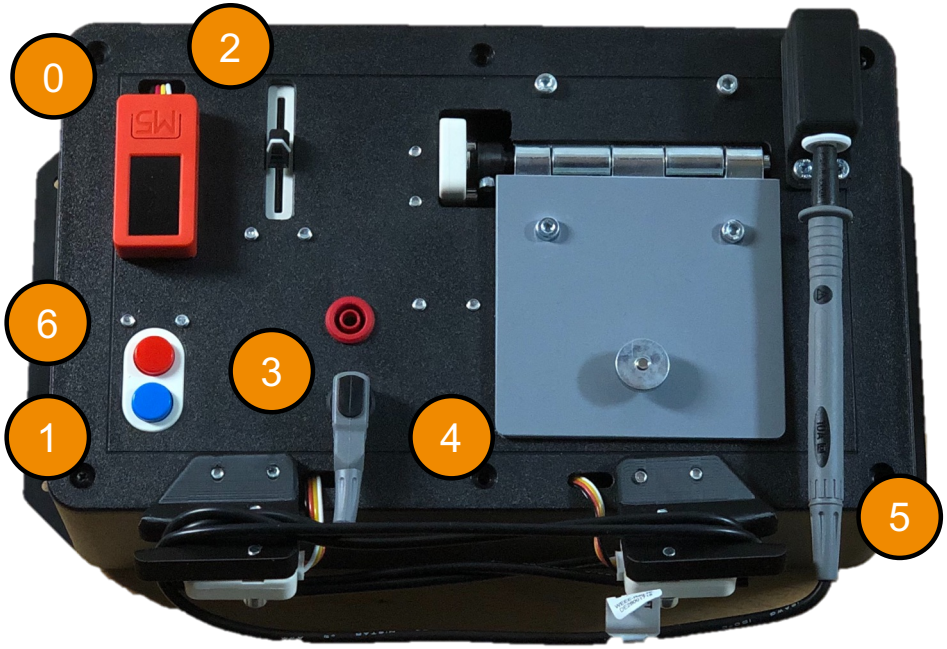
**Develop Autonomous Robot Skills to:**

- Locate / Recognize Object
- Perceive & Manipulate Instruments
- Plug In Probe
- Access & Probe Circuit
- Clean Up Workspace



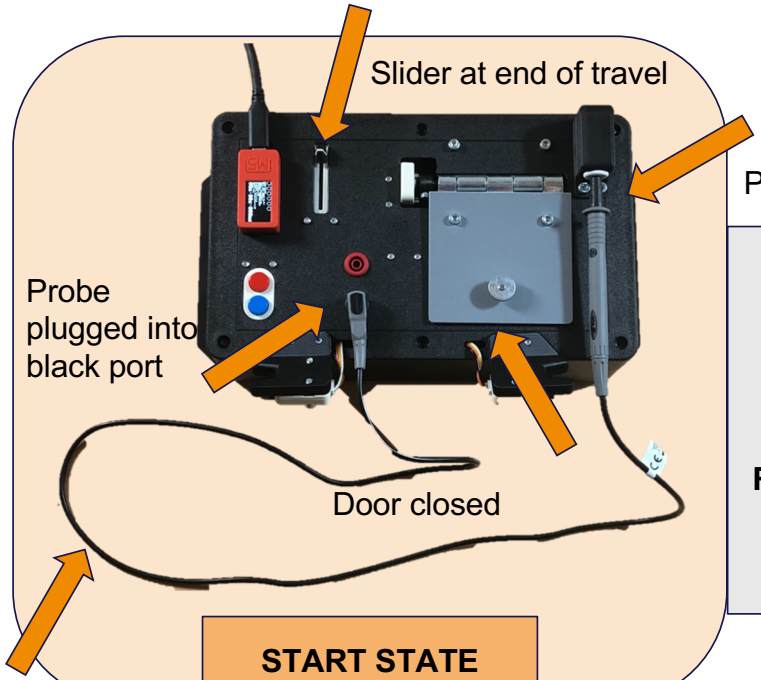
# Robothon® 2023 – Task Board & Trial Protocol

0. Press Start Trial Button
1. Find Board & Press Button
2. Move Slider to Setpoints on Screen
3. Plug in Probe into Test Port
4. Open Door, Probe Circuit
5. Wrap Cable, Replace Probe
6. Press Stop Trial Button




Task Board is fixed on flat table with velcro strips and repositioned before each trial.

# Task Board Tasks: Start and Goal States



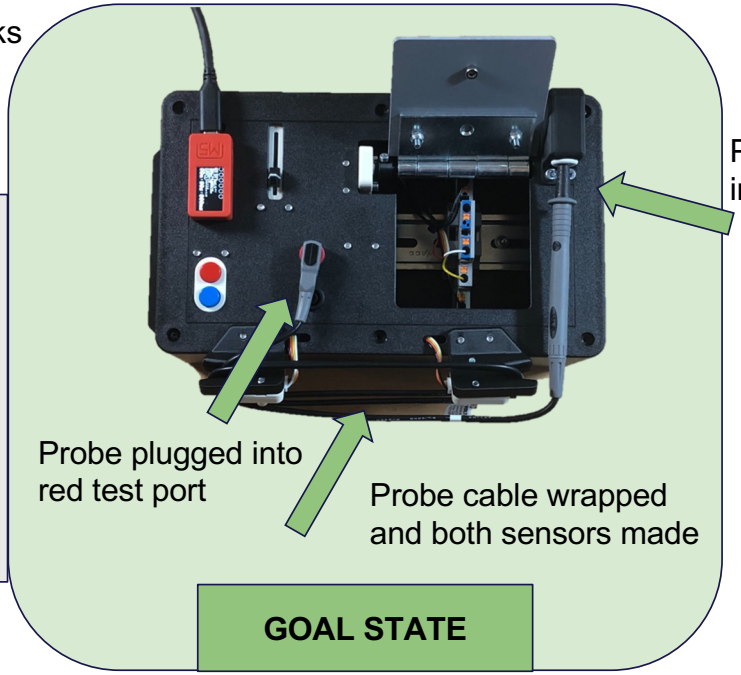

All subtasks completed

**Random Start Position.**

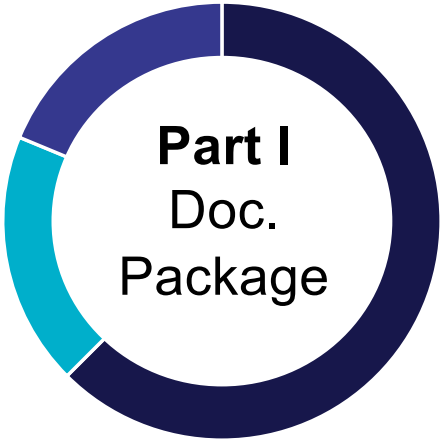
**10-minutes.**

**Robot Platform only.**

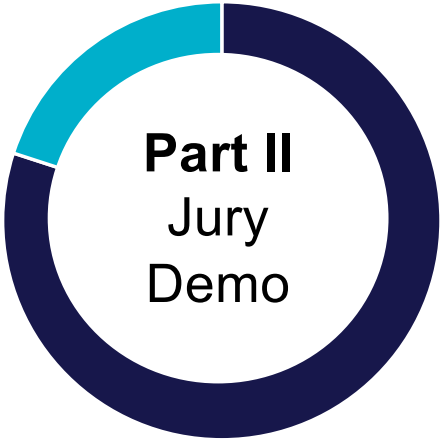
**No assistance.**



# Team Evaluations




Team Video
Robustness Video
Repository




Point Score
(Part I + II)



Task Board
Execution
Time

# Team Evaluations

| | Score Category | Description | Max Value |
|---|---|---|---|
| Part I | 5x Back-to-Back Trials with Task Board Video Recording | Continuous video recording (max length 10min) with points awarded from task board Bonus for ***Robustness***. | 20 x 5 = 100 |
| | Solution Transferability Demo | Team includes video of their robot working with a Bring-Your-Own-Device "BYOD" object, representative of local e-waste, e.g., smoke detector, remote, etc. | 30 |
| | Project Documentation | Team provides accurate description of their robot platform, software dependencies and instructions on how to start their system in a public repository. | 30 |
| Part II | Live Presentation | Team demonstrates solution to task board and transferability demo during 30min Zoom call. Bonus for ***Flexibility*** | 20 + 20 = 40 |
| | **Total Team Score** | ****Finalists selected by Grand Jury*** | **200** |